Spencer Hilligoss

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Education

PhD in Statistics *Sep 2023 – Present*
University of California, Irvine *(Expected December 2027)*

- **Coursework:** Advanced Probability, Machine Learning, Statistical Consulting
- **Research Topics**: Time-series Forecasting, Deep Learning, Generative Modeling, Causal Inference
- **Supervised by:** Dr. Annie Qu and Dr. Tianchen Qian
- **GPA:** 3.929/4.0

B.S. in Statistics and Operations Research, B.A. in Economics *Aug 2019 – May 2023*
University of North Carolina at Chapel Hill

- **Coursework:** Probability, Statistical Machine Learning, Real Analysis, Stochastic Modeling, Dynamic Decision Making, Financial Markets, Micro and Macro Economic Theory
- **GPA:** 3.7/4.0

Fellowships and Awards

University of California, Irvine

- T32 STEER in Biomedical Sciences Fellow *Aug 2025 - Present*
- Diversity Recruitment Fellowship *Aug 2023*

Publications and Current Work

Hilligoss, S., Qu, A. (2026). "Duet-TS: Dual-Pathway Diffusion for Heterogeneous Time Series." *Under review, NeurIPS 2026.*

Hilligoss, S., Qu, A. (2026). "Causal Mediation Pathways in Continuous Postprandial Glucose Monitoring for Type 1 Diabetes Patients." *Accepted, npj Metabolic Health and Disease.*

Hilligoss, S., Saha, A., Qian, T. "Selective Inference for Time-Varying Causal Effect Moderation via the Ordered Lasso." *Manuscript in preparation.*

Research and Work Experience

Duet-TS: Dual-Pathway Diffusion for Heterogeneous Time Series *Mar 2026 – Aug 2026*
University of California, Irvine — under review, NeurIPS 2026

- Led development of a dual-pathway diffusion model for heterogeneous time-series generation, mirroring the mixed-effects decomposition $y = X\beta + Zb$; generated samples matched subject-level distributions more closely than the PaD-TS baseline on every benchmark (clinical glucose, mobile sensing, equities).
- Validated results in peer review with multi-seed confidence intervals, ablations, and out-of-sample checks.

Sentiment-Enhanced LSTM Forecasting (DJIA) *Sep 2024 - Dec 2024*
University of California, Irvine

- Engineered a daily feature set combining market data with text-based sentiment signals (news/social text) and aligned features to next-day index returns.
- Implemented and trained LSTM forecasting models; compared against baseline time-series models using rolling/expanding-window backtests and out-of-sample error metrics.

The Lasting Impact of HOLC Redlining on Minorities in New York *June 2022 - Aug 2022*
Harvard University, Research and Policy Institute

- Built a reproducible R workflow linking HOLC redlining maps to demographic data; estimated regression-based effects of historical redlining on present-day outcomes with robustness and sensitivity checks.

Mentorship

Undergraduate Research Mentor, University of California, Santa Barbara

- Weekly research meetings with two undergraduates: financial time-series modeling and generative diffusion modeling applied to soccer tactics.

Computing Tools

Python (PyTorch, NumPy, Pandas, scikit-learn); R; Git; Linux; HPC/Slurm; Stata; Excel